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(54) **ACOUSTIC WAVE FILTER AND RADIO FREQUENCY MODULE**

(71) Applicant: **MURATA MANUFACTURING CO., LTD.**, Kyoto (JP)

(72) Inventor: **Shun HARADA**, Kyoto (JP)

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CPC ***H03H 9/6483*** (2013.01); ***H03F 1/26*** (2013.01); *H03F 2200/294* (2013.01)

(57) **ABSTRACT**

An acoustic wave filter includes: a plurality of serial arm resonant devices including serial arm resonators disposed on a serial arm path connecting input/output terminals; one or more parallel arm resonant devices including a parallel arm resonator connected between the serial arm path and the ground; an inductor connected between the ground and a first path connecting the serial arm resonators. The serial arm resonator is connected closest to the input/output terminal among the plurality of serial arm resonant devices, the one or more parallel arm resonant devices, and the inductor, and the serial arm resonator has the highest resonant frequency among the plurality of serial arm resonant devices.

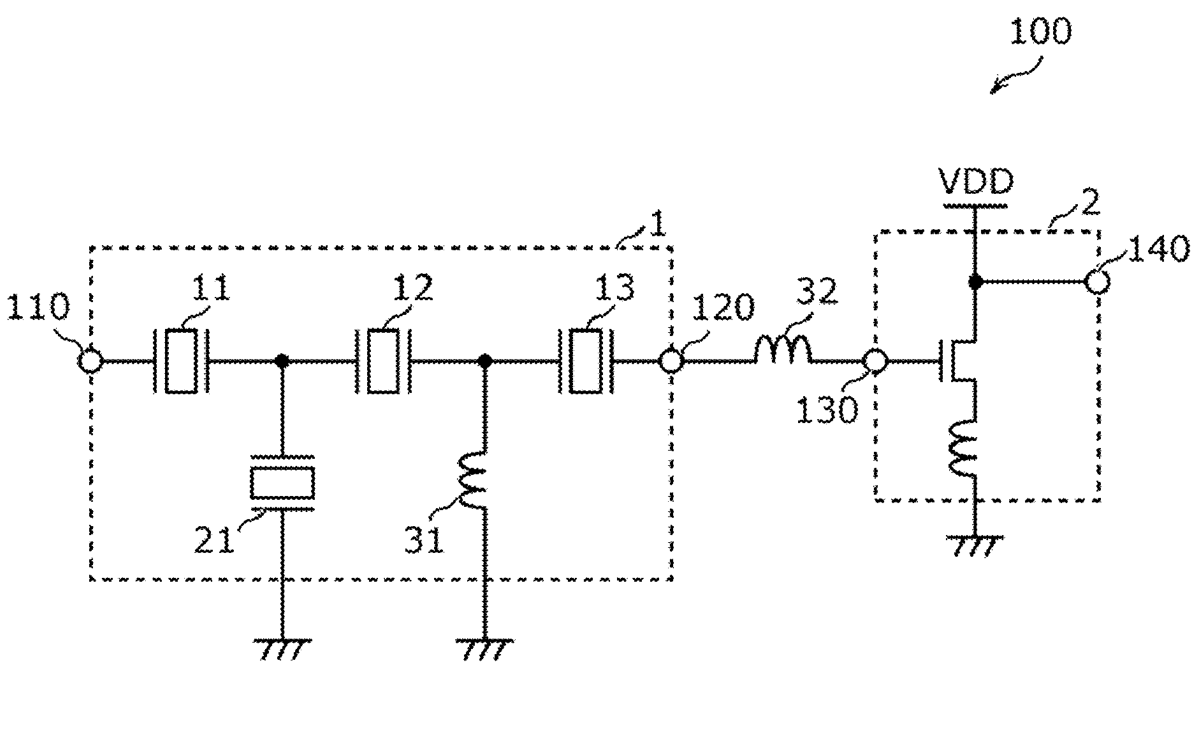
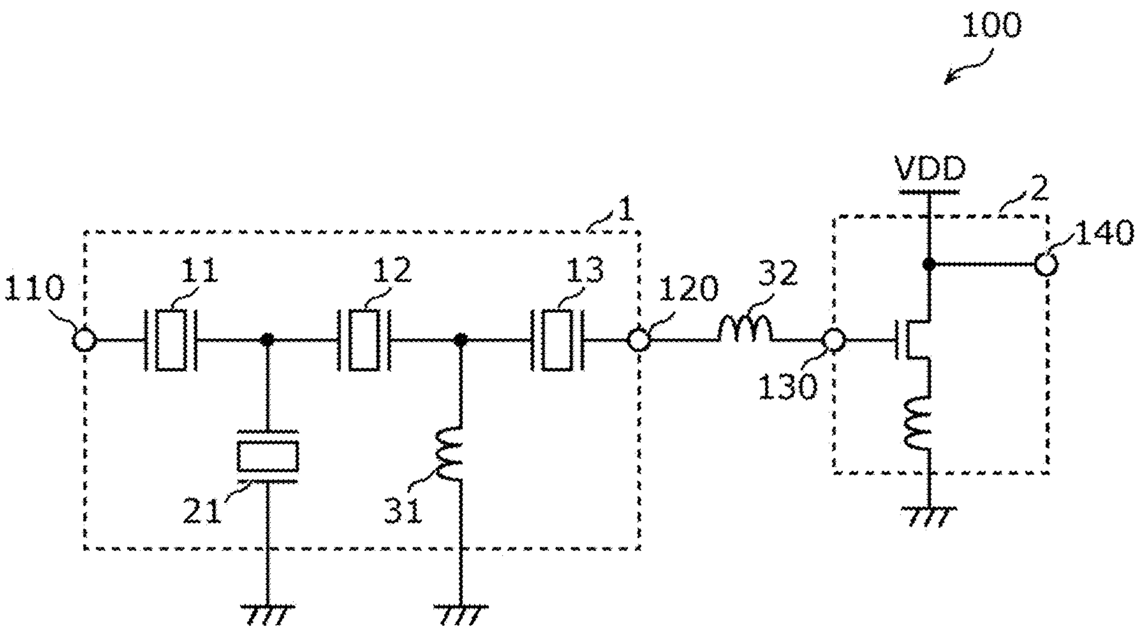


FIG. 1



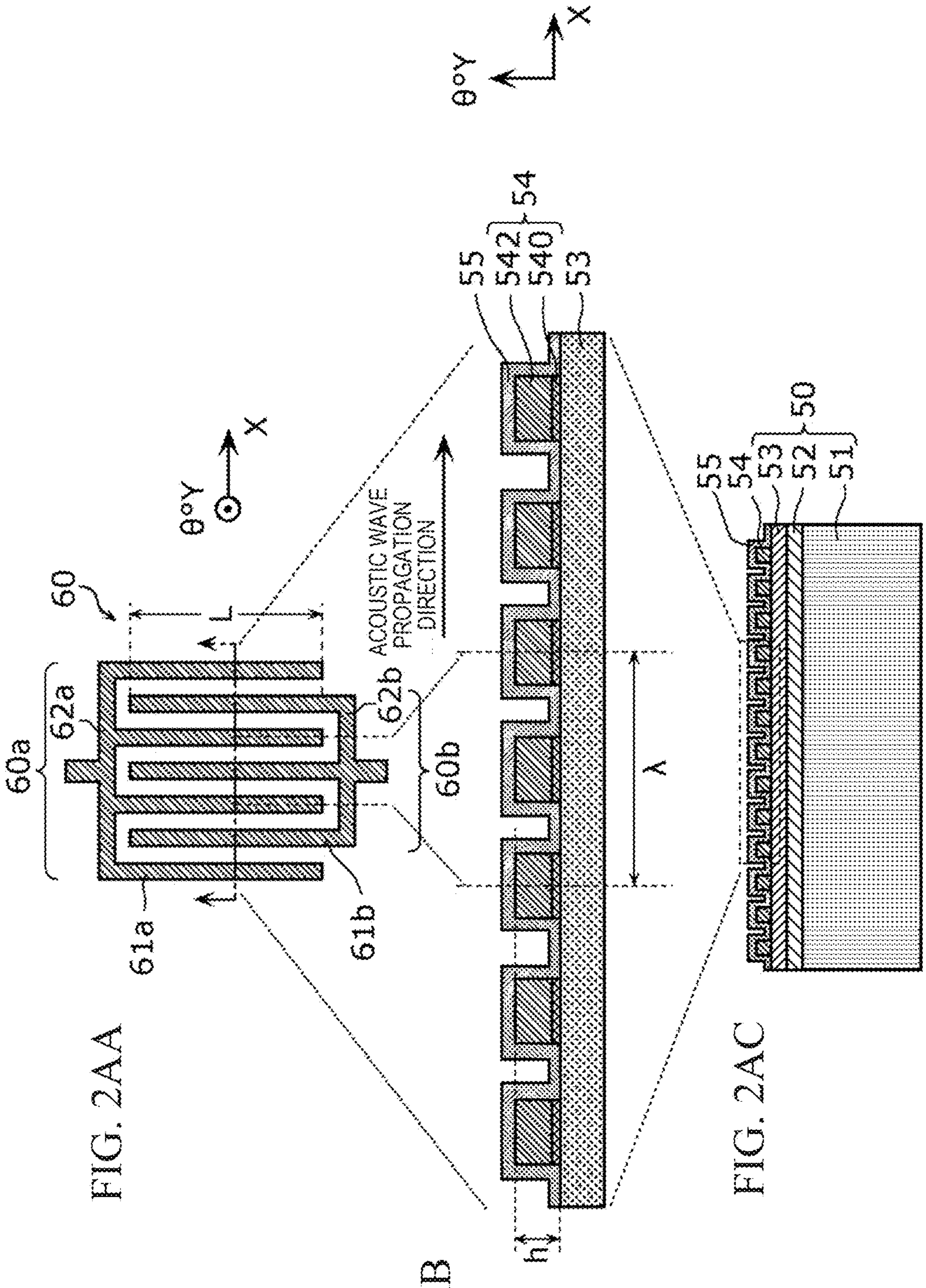


FIG. 2AA

FIG. 2AB

FIG. 2AC

FIG. 2B

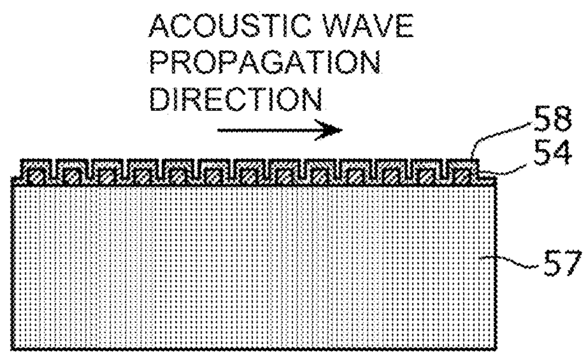


FIG. 2C

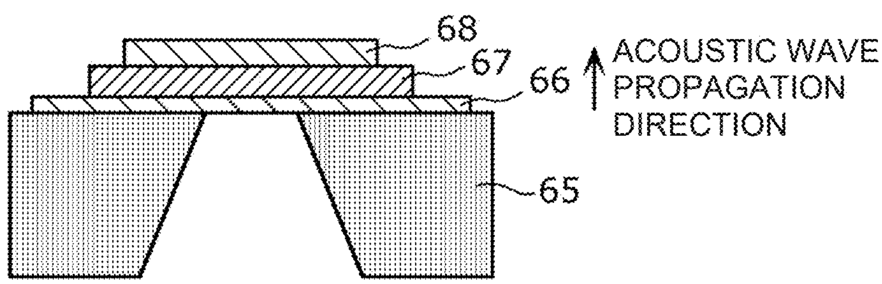


FIG. 3A

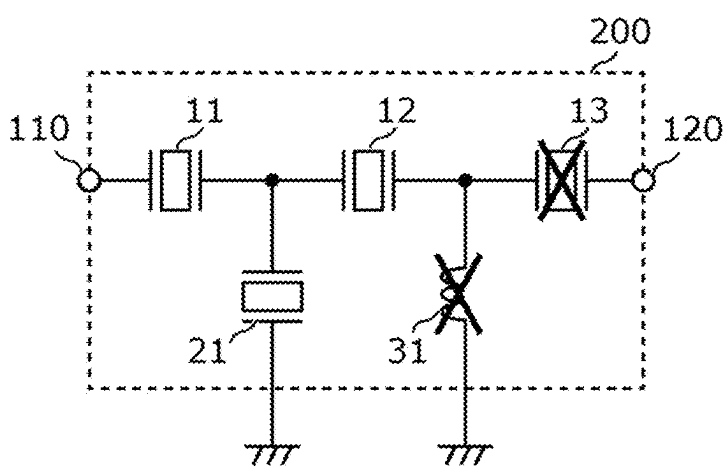


FIG. 3BA

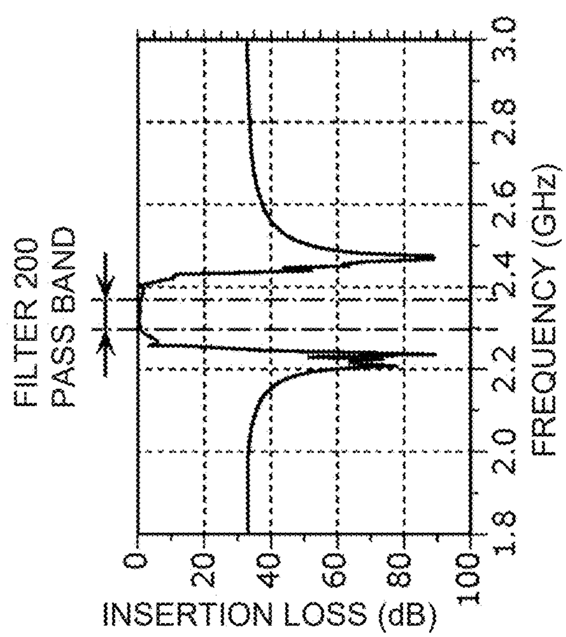


FIG. 3BB

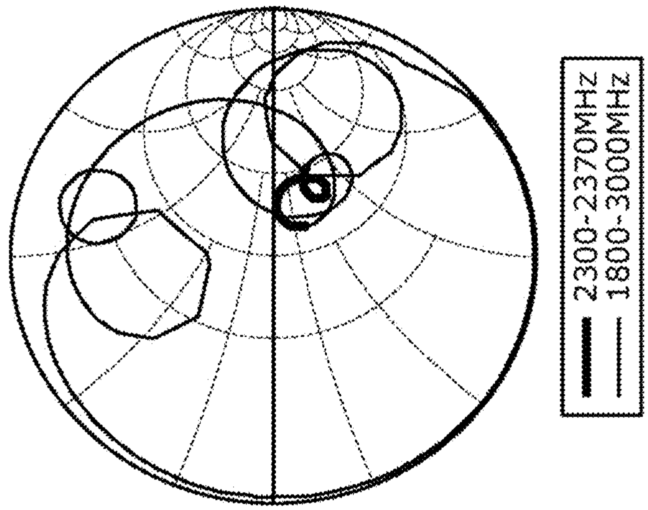


FIG. 4A

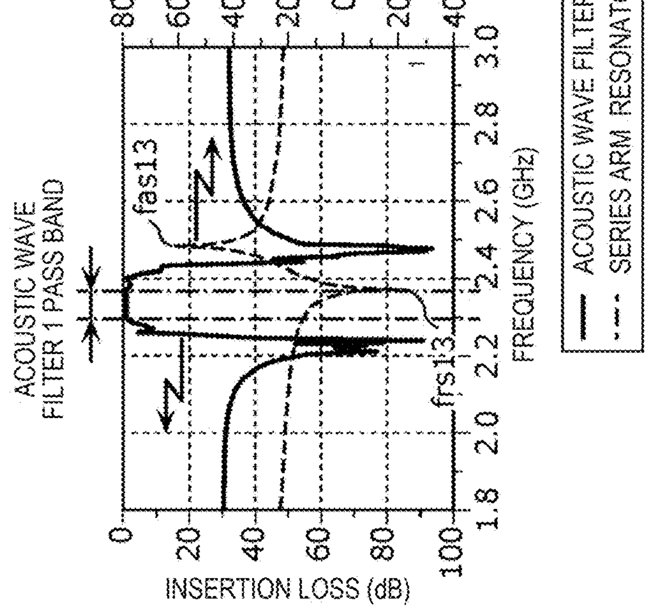


FIG. 4B

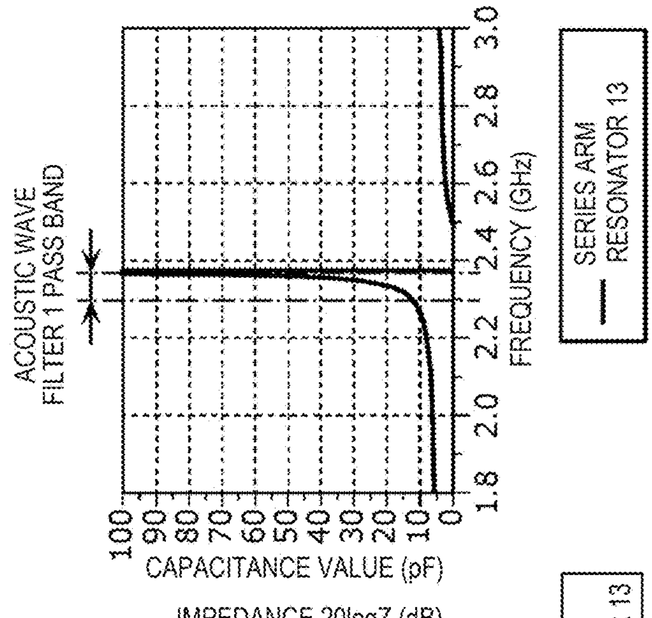


FIG. 4C

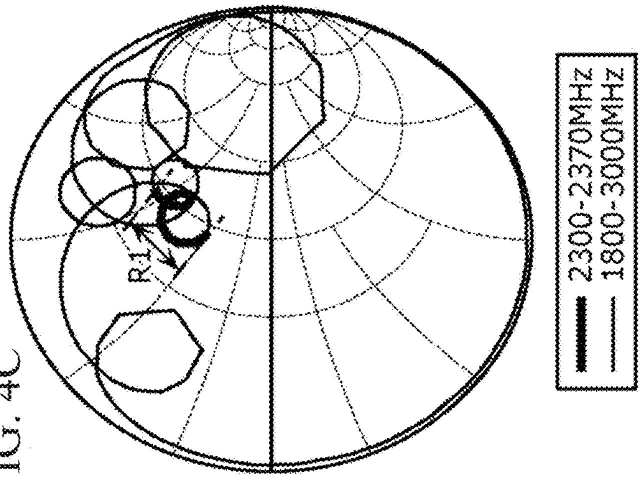


FIG. 5C

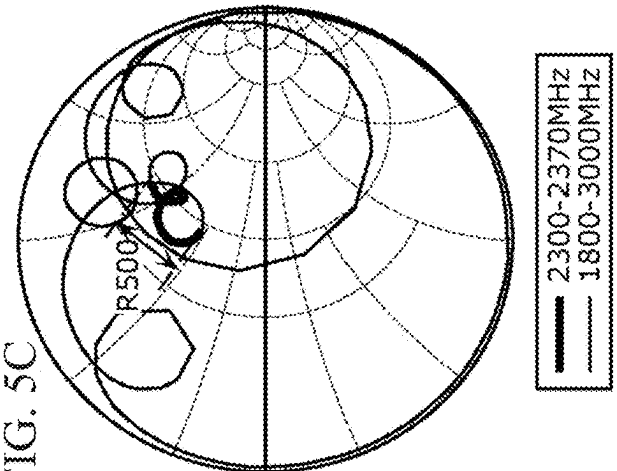


FIG. 5B

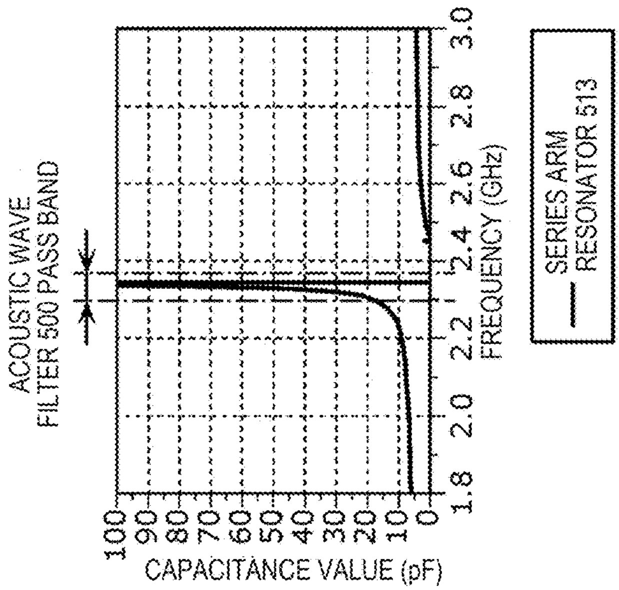


FIG. 5A

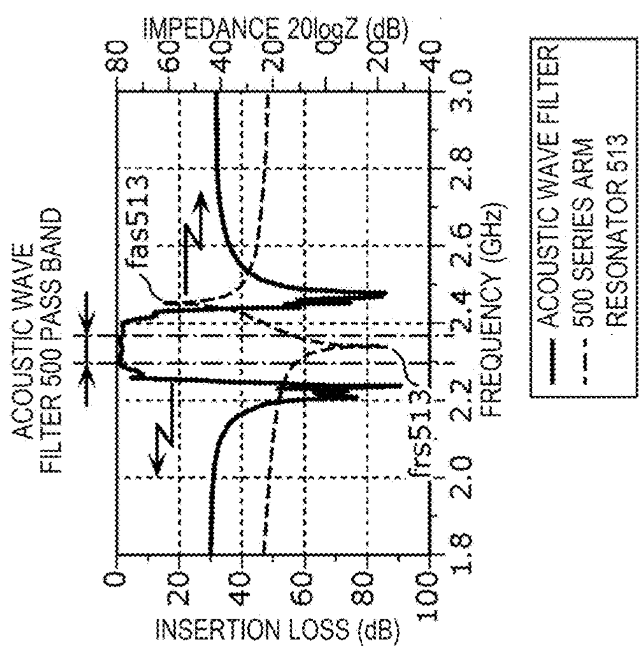


FIG. 6A

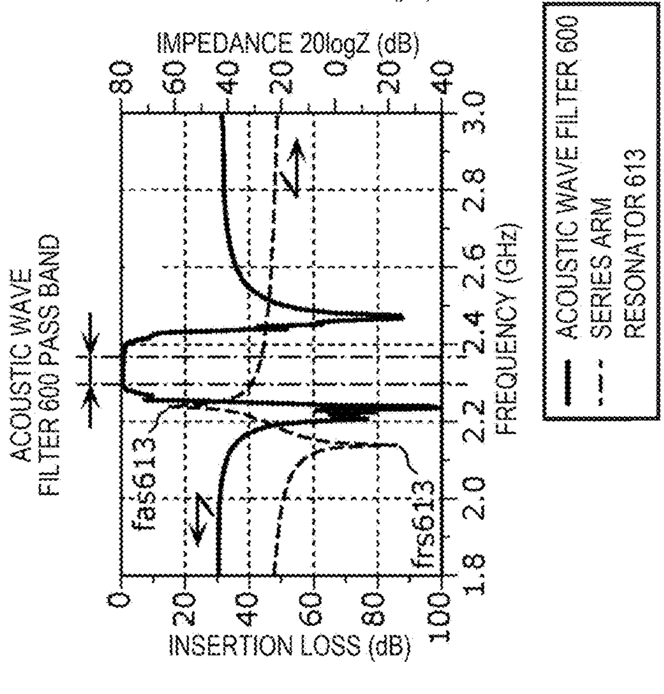


FIG. 6B

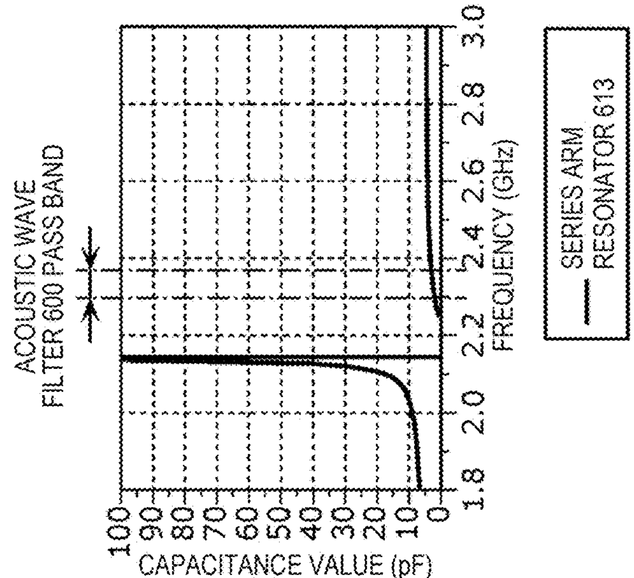


FIG. 6C

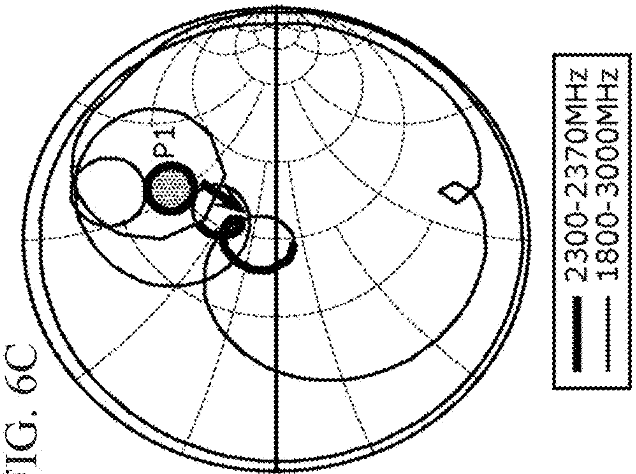


FIG. 7A

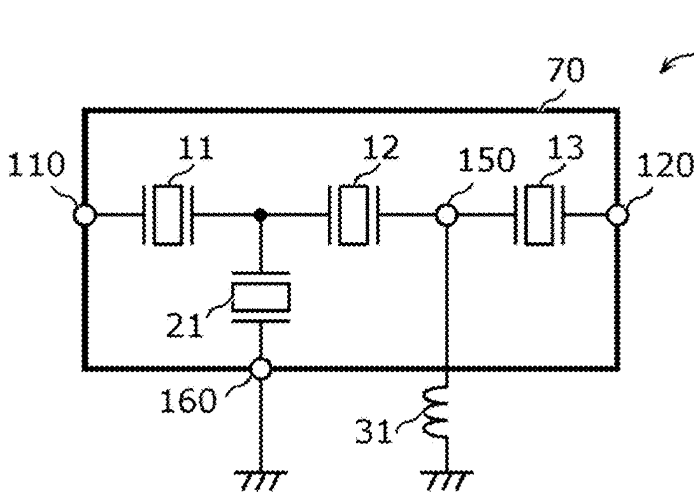


FIG. 7B

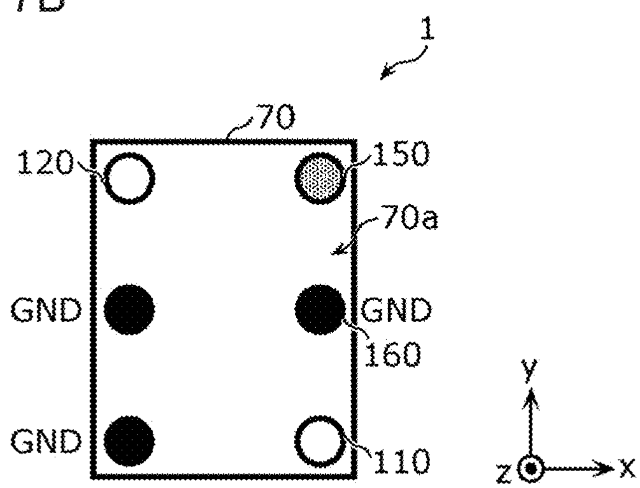


FIG. 7C

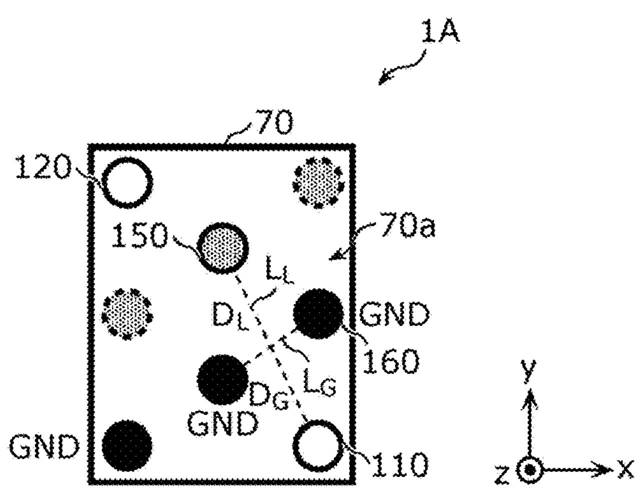


FIG. 8A

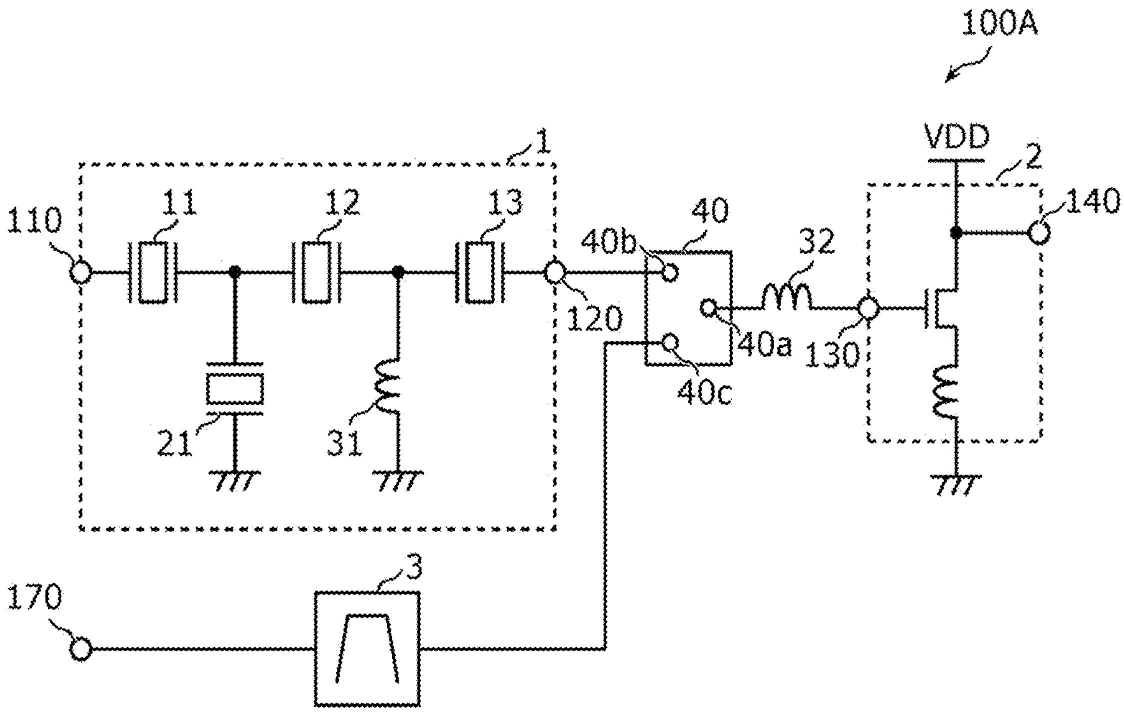
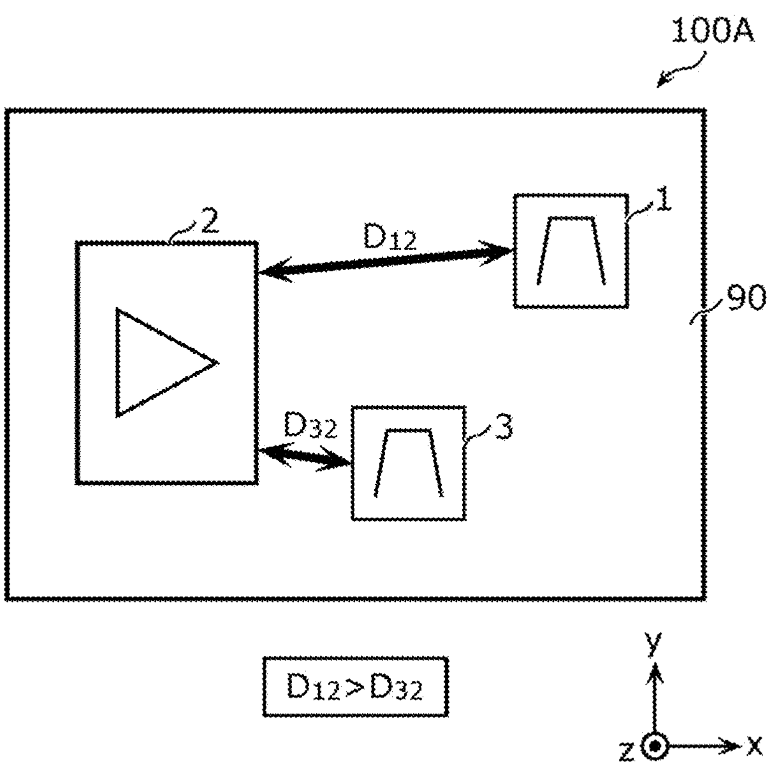

100A
VDD
1
2
110
11
12
13
40
40b
32
140
120
40a
130
40c
21
31
170
3

FIG. 8B


100A
2
D_{12}
1
90
D_{32}
3
$D_{12}>D_{32}$
y
z
x

ACOUSTIC WAVE FILTER AND RADIO FREQUENCY MODULE

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority from Japanese Patent Application No. 2024-026594 filed on Feb. 26, 2024. The content of this application is incorporated herein by reference in its entirety.

BACKGROUND ART

[0002] The present disclosure relates to an acoustic wave filter and a radio frequency module.

[0003] International Publication No. 2019/064990 (FIG. **13**) discloses a radio frequency front end circuit including low-noise amplifiers that amplify radio frequency reception signals and bandpass filters connected to the input terminals of the respective low-noise amplifiers.

BRIEF SUMMARY

[0004] Each low-noise amplifier has capacitive input impedance. To achieve impedance matching between the low-noise amplifier and a corresponding one of bandpass filter in the radio frequency front end circuit disclosed, for example, in International Publication No. 2019/064990, it is effective to connect an inductor between the ground and a path connecting the low-noise amplifier and the bandpass filter. However, connecting the inductor needs to dispose a DC blocking capacitor in series to the path to prevent leakage of DC bias current to be supplied to the low-noise amplifier.

[0005] However, if the DC blocking capacitor is disposed in series to the input terminal of the low-noise amplifier, impedance mismatching occurs in the frequency band for a radio frequency reception signal, leading to the occurrence of transmission loss. As the result, the noise figure of the low-noise amplifier is deteriorated on occasions.

[0006] Accordingly, the present disclosure provides an acoustic wave filter and a radio frequency module with which the DC blocking and the impedance matching in the pass band are ensured.

[0007] An acoustic wave filter according to an aspect of the present disclosure includes a plurality of serial arm resonant devices including a first serial arm resonant device and a second serial arm resonant device, the first serial arm resonant device being disposed on a serial arm path connecting a first input/output terminal and a second input/output terminal; one or more parallel arm resonant devices connected between the serial arm path and ground; and a first inductor connected between the ground and a first path connecting the first serial arm resonant device and the second serial arm resonant device. The first serial arm resonant device is connected closest to the first input/output terminal among the plurality of serial arm resonant devices, the one or more parallel arm resonant devices, and the first inductor, and the first serial arm resonant device has a highest resonant frequency among the plurality of serial arm resonant devices.

[0008] A radio frequency module according to an aspect of the present disclosure includes the acoustic wave filter and the low-noise amplifier having the input terminal connected to the first input/output terminal.

[0009] According to the present disclosure, the acoustic wave filter and the radio frequency module with which the DC blocking and the impedance matching in the pass band are ensured may be provided.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a view of the circuit configuration of an acoustic wave filter and a radio frequency module according to an embodiment;

[0011] FIGS. **2**AA-**2**AC depict a plan view and a cross-sectional view schematically illustrating a first example of an acoustic wave resonator included in the acoustic wave filter according to the embodiment;

[0012] FIG. **2**B is a cross-sectional view schematically illustrating a second example of the acoustic wave resonator included in the acoustic wave filter according to the embodiment;

[0013] FIG. **2**C is a cross-sectional view schematically illustrating a third example of the acoustic wave resonator included in the acoustic wave filter according to the embodiment;

[0014] FIG. **3**A is a view of the circuit configuration of an acoustic wave filter according to Comparative Example 1;

[0015] FIGS. **3**BA and **3**BB illustrate a Smith chart representing the bandpass characteristic and the impedance of the acoustic wave filter according to Comparative Example 1;

[0016] FIGS. **4**A-**4**C illustrate a Smith chart representing the bandpass characteristic of the acoustic wave filter according to the embodiment, the capacitance characteristic of a first serial arm resonant device, and the impedance of the acoustic wave filter;

[0017] FIGS. **5**A-**5**C illustrate a Smith chart representing the bandpass characteristic of an acoustic wave filter according to Comparative Example 2, the capacitance characteristic of a first serial arm resonant device, and the impedance of the acoustic wave filter;

[0018] FIGS. **6**A-**6**C illustrate a Smith chart representing the bandpass characteristic of an acoustic wave filter according to Comparative Example 3, the capacitance characteristic of a first serial arm resonant device, and the impedance of the acoustic wave filter;

[0019] FIG. **7**A is a view of the circuit configuration of the acoustic wave filter according to the embodiment;

[0020] FIG. **7**B is a plan view of the terminal layout of the acoustic wave filter according to the embodiment;

[0021] FIG. **7**C is a plan view of the terminal layout of an acoustic wave filter according to Modification 1 of the embodiment;

[0022] FIG. **8**A is a view of the circuit configuration of a radio frequency module according to Modification 2 of the embodiment; and

[0023] FIG. **8**B is a plan view of the component configuration of the radio frequency module according to Modification 2 of the embodiment.

DETAILED DESCRIPTION

[0024] Hereinafter, an embodiment of the present disclosure will be described in detail by using the drawings. The embodiment to be described later represents a comprehensive or specific example. Accordingly, a numerical value, a shape, a material, a component, the arrangement of the component and the like described in the following embodiment is an example and are not intended to limit the present disclosure. Among components in the following embodiment, a component that is not described in an independent claim is described as an optional component. The sizes and the ratio of the sizes of components in the drawings are not necessarily precisely illustrated.

[0025] Each drawing is a schematic view appropriately subjected to emphasis, omission, or ratio control to describe the present disclosure, is not necessarily strictly illustrated, and has a shape, a positional relationship and a ratio different from actual ones on occasions. Substantially the same components are denoted by the same reference numerals throughout the drawings and redundancy is omitted or simplified in some cases.

[0026] In the circuit configuration of the present disclosure, the term "connected" includes not only "directly connected" by using a connection terminal and/or a wiring conductor but also "electrically connected" with a matching element or a switching circuit interposed between one component and the other component". In addition, the phrase "connected between A and B" denotes that a component is connected between A and B and to both of A and B.

[0027] In the present disclosure, the term "terminal" denotes a point where a conductor in the element ends. If the impedance of the conductor between elements is sufficiently low, the terminal is interpreted as not only a single point but also as any point (a node) on the conductor between the elements or the entire conductor.

[0028] In the circuit element layout of the present disclosure, the phrase "a circuit element A is disposed in series to a path B" denotes that the signal input terminal and the signal output terminal of the circuit element A are connected to two respective wiring elements forming at least part of the path B. At least one of the two wiring elements may be an electrode or a terminal.

[0029] In the drawings described below, an x axis and a y axis are axes orthogonal to each other on a plane parallel to the main surface of a substrate. Specifically, if the substrate has a rectangular shape in plan view, the x axis is parallel to a first side of the substrate, and the y axis is parallel to a second side orthogonal to the first side of the substrate. A z axis is an axis perpendicular to the main surface of the substrate. The positive direction of the z axis denotes an upward direction, and the negative direction thereof denotes a downward direction.

[0030] Terms representing a relationship between elements such as "parallel" and "perpendicular", a term representing the shape of an element such as "rectangular", and a numerical value range indicate not only strict meaning but also inclusion of substantially the same range, for example, an error of approximately several percent.

[0031] In the component layout of the present disclosure, the phrase "in plan view of the main surface (of the substrate)" denotes seeing an object orthographically projected on the xy plane in the positive direction of the z axis. The phrase "A overlaps with B in plan view" denotes that at least part of the region of A orthographically projected on the xy plane overlaps with at least part of the region of B orthographically projected on the xy plane. The phrase "A is disposed between B and C" denotes that at least one of a plurality of lines connecting any point in B and any point in C passes through A.

[0032] In the component layout of the present disclosure, the phrase "a component is disposed on the substrate" includes disposition of the component on the main surface of the substrate and disposition of the component in the substrate. The phrase "a component is disposed on the main surface of a substrate" includes disposition of the component above the main surface without necessarily being in contact with the main surface (for example, the component is stacked on a different component disposed in contact with the main surface), in addition to disposition of the component in contact with the main surface of the substrate. The phrase "a component is disposed on the main surface of a substrate" may also include disposition of the component in a recessed portion formed in the main surface. The term "a component is disposed in the substrate" includes: encapsulation of the component in the module substrate; disposition of the entire component between main surfaces of the substrate but exposure of part of the component from the substrate; and disposition of only part of the component in the substrate.

[0033] In the embodiment below, the pass band of a filter is defined as a frequency band between two frequencies one of which has a value of 3 dB higher than the lowest value of insertion loss in the pass band.

[0034] The term "acoustic wave resonant device (a serial arm resonant device and a parallel arm resonant device) is defined as one of: (1) a resonance circuit including an acoustic wave resonator and a circuit connected in parallel to the acoustic wave resonator (or a circuit element) (a circuit in parallel connection between an acoustic wave resonator and a circuit (or a circuit element); (2) a resonance circuit (a circuit in serial connection between an acoustic wave resonator and a circuit (or a circuit element)) having a configuration that includes a circuit (or a circuit element) connected to an acoustic wave resonator and only one of two input/output terminals of the acoustic wave resonator and in which a different circuit (and a different circuit element) and the ground are not connected to a connection node connecting the acoustic wave resonator and the circuit (or the circuit element); (3) a resonance circuit (a circuit in parallel connection between divided resonators) including a plurality of acoustic wave resonators connected in parallel to each other; and (4) a resonance circuit (a circuit in series connection between divided resonators) that includes a plurality of acoustic wave resonators connected in series to each other and that has a configuration in which a circuit (and a circuit element) other than the plurality of acoustic wave resonators and the ground are not connected to a connection node connecting a plurality of acoustic wave resonators.

[0035] The resonant frequency and the anti-resonant frequency described in the above-mentioned embodiment and modifications are derived in the following manner. For example, in a state where the acoustic wave resonator or the acoustic wave resonant device is not connected to a different circuit element, a RF probe is brought into contact with two input/output electrodes of the acoustic wave resonator or the acoustic wave resonant device, and reflection characteristics (impedance characteristics) are measured with a network analyzer or the like.

[0036] The electrostatic capacitance of the acoustic wave resonator or the acoustic wave resonant device is derived in the following manner. In a state where the acoustic wave resonator or the acoustic wave resonant device is not connected to a different circuit element, a probe is brought into contact with two input/output electrodes of the acoustic wave resonator or the acoustic wave resonant device, and capacitance in DC is measured with a network analyzer, an impedance meter, or the like. The electrostatic capacitance may be compared between a plurality of surface acoustic wave resonators or between a plurality of surface acoustic wave resonant devices by actually measuring the number of pairs, the interdigital width, the pitch, and the duty of interdigital transducer (IDT) electrodes.

[0037] In the present disclosure, the term "band" denotes at least one of an uplink operating band and a downlink operating band in a frequency band defined in advance, for a communication system built up by using, for example, radio access technology (RAT), by a standardization organization or the like (such as 3GPP (registered trademark) or Institute of Electrical and Electronics Engineers (IEEE)). In this embodiment, for example, a long term evolution (LTE) system, a 5th Generation—New Radio (5G-NR) system, a wireless local area network (WLAN) system, and the like are usable as a communication system, but the communication system is not limited to these. The uplink operating band in the frequency band denotes a frequency range designated for an uplink in the frequency band. The downlink operating band in the frequency band denotes a frequency range designated for a downlink.

EMBODIMENT

1 Circuit Configuration of Acoustic Wave Filter 1 and Radio Frequency Module 100

[0038] FIG. **1** is a view of the circuit configuration of an acoustic wave filter **1** and a radio frequency module **100** according to an embodiment. As illustrated in FIG. **1**, the radio frequency module **100** includes the acoustic wave filter **1**, a low-noise amplifier **2**, and an inductor **32**.

[0039] The low-noise amplifier **2** has an input terminal **130** and an output terminal **140**, and the input terminal **130** is connected to an input/output terminal **120** of the acoustic wave filter **1** with the inductor **32** interposed therebetween. The low-noise amplifier **2** includes an amplifying transistor that is, for example, a field effect transistor (FET) or a bipolar transistor (Bipolar Transistor). The gate (or the base) of the amplifying transistor is connected to the input terminal **130** without necessarily a capacitive element interposed therebetween, the drain (or the collector) thereof is connected to the output terminal **140**, and the source (or the emitter) thereof is connected to the ground. Further, a DC bias voltage (DC bias current) is supplied to the gate (or the base) of the amplifying transistor. With the configuration above, in response to the supply of the DC bias voltage (DC bias current) to the gate (or the base), the low-noise amplifier **2** thereby amplifies a radio frequency signal having passed through the acoustic wave filter **1** and outputs the radio frequency signal to the output terminal **140**. The input impedance of the low-noise amplifier **2** is capacitive and high impedance.

[0040] The inductor **32** is an example of a second inductor and is disposed in series to a path connecting the input/output terminal **120** and the input terminal **130**. The inductor **32** is a circuit element for performing impedance matching between the low-noise amplifier **2** having the capacitive input impedance and the acoustic wave filter **1**. The inductor **32** does not have to be provided.

[0041] The acoustic wave filter **1** is a bandpass filter and includes serial arm resonators **11**, **12**, and **13**, a parallel arm resonator **21**, an inductor **31**, an input/output terminal **110**, and the input/output terminal **120**.

[0042] Each of the serial arm resonators **11** to **13** is an example of the acoustic wave resonant device including an acoustic wave resonator and is disposed on a serial arm path connecting the input/output terminal **110** (a second input/output terminal) and the input/output terminal **120** (a first input/output terminal). The serial arm resonators **11** to **13** are formed as a plurality of serial arm resonant devices included in the acoustic wave filter **1**.

[0043] One serial arm resonant device (acoustic wave resonant device) is composed of the serial arm resonator **11** only. The serial arm resonator **12** is an example of a second serial arm resonant device, and one serial arm resonant device (acoustic wave resonant device) is composed of the serial arm resonator **12** only. The serial arm resonator **13** is an example of a first serial arm resonant device, and one serial arm resonant device (acoustic wave resonant device) is composed of the serial arm resonator **13** only. The serial arm resonators **11** to **13** are connected in the order of the serial arm resonators **11**, **12**, and **13** from the input/output terminal **110**.

[0044] The parallel arm resonator **21** is an example of an acoustic wave resonant device including an acoustic wave resonator and is connected between the serial arm path and the ground. One or more parallel arm resonant devices included in the acoustic wave filter **1** are composed of the one or more parallel arm resonators **21**. The parallel arm resonator **21** is connected between the ground and a point of connection between the serial arm resonators **11** and **12**. One parallel arm resonant device (acoustic wave resonant device) is composed of the parallel arm resonator **21** only.

[0045] The inductor **31** is an example of a first inductor and is connected between the ground and a first path connecting the serial arm resonators **12** and **13**.

[0046] The serial arm resonator **13** is connected closest to the input/output terminal **120** among the serial arm resonators **11** to **13**, the parallel arm resonator **21**, and the inductor **31**.

[0047] The serial arm resonator **13** also has the highest electrostatic capacitance among the serial arm resonators **11** to **13** and the parallel arm resonator **21** and has the highest resonant frequency among the serial arm resonators **11** to **13**.

[0048] The acoustic wave filter **1** according to this embodiment is only required to include at least the serial arm resonators **12** and **13** as the plurality of serial arm resonant devices and may include four or more serial arm resonant devices including the serial arm resonators **12** and **13**. The acoustic wave filter **1** according to this embodiment is only required to include at least the parallel arm resonator **21** as the one or more parallel arm resonant devices and may include two or more parallel arm resonant devices including the parallel arm resonator **21**. In addition, in the acoustic wave filter **1** according to this embodiment, at least one of an inductor and a capacitor may be disposed in series on the serial arm path connecting the input/output terminal **110** and the serial arm resonator **12**, and at least one of an inductor and a capacitor may be connected between the serial arm path and the ground. An inductor may be connected between the parallel arm resonator **21** and the ground.

[0049] Each of the serial arm resonators **11** to **13** and the parallel arm resonator **21** (acoustic wave resonant devices) has only one acoustic wave resonator but may be one of, for example, (1) a resonant device composed of a circuit including an acoustic wave resonator and at least one of a capacitor and an inductor connected in parallel to the acoustic wave resonator; (2) a resonant device composed of a circuit including an acoustic wave resonator and at least one of a capacitor and an inductor connected in series to the acoustic wave resonator; (3) a resonant device composed of a plurality of acoustic wave resonators connected in parallel; and (4) a resonant device composed of a plurality of acoustic wave resonators connected in series.

[0050] The acoustic wave filter **1** according to this embodiment may also include a longitudinally coupled resonant device in addition to the serial arm resonant devices and the parallel arm resonant device included in the ladder filter.

2 Structure of Acoustic Wave Resonator

[0051] The structure of the acoustic wave resonators (serial arm resonators and the parallel arm resonator) included in the acoustic wave filter **1** will then be exemplified.

[0052] FIGS. **2**AA-**2**AC depict a plan view and a cross-sectional view schematically illustrating a first example of the acoustic wave resonators included in the acoustic wave filter **1** according to the embodiment. FIGS. **2**AA-**2**AC illustrate the basic structure of each of the plurality of acoustic wave resonators included in the acoustic wave filter **1**. A surface acoustic wave resonator **60** illustrated in FIGS. **2**AA-**2**AC is provided to explain a typical structure of the surface acoustic wave resonator included in the acoustic wave filter **1**, and the number, the length, and the like of electrode fingers included in the electrode are not limited to these.

[0053] The surface acoustic wave resonator **60** includes a piezoelectric substrate **50** and comb-shaped electrodes **60***a* and **60***b*.

[0054] As illustrated in FIG. **2**AA, a pair of comb-shaped electrodes **60***a* and **60***b* facing each other are formed on the piezoelectric substrate **50**. The comb-shaped electrode **60***a* includes a plurality of electrode fingers **61***a* parallel to each other and a busbar electrode **62***a* connecting the plurality of electrode fingers **61***a*. The comb-shaped electrode **60***b* includes a plurality of electrode fingers **61***b* parallel to each other and a busbar electrode **62***b* connecting the plurality of electrode fingers **61***b*. The plurality of electrode fingers **61***a* and **61***b* are formed in a direction orthogonal to an acoustic wave propagation direction (X-axis direction).

[0055] An IDT electrode **54** including the plurality of electrode fingers **61***a* and **61***b* and the busbar electrodes **62***a* and **62***b* has a layered structure having a close-contact layer **540** and a main electrode layer **542**, as illustrated in FIG. **2**AB.

[0056] The close-contact layer **540** is a layer for improving close contact between the piezoelectric substrate **50** and the main electrode layer **542**, and for example, Ti is used as a material thereof. As a material of the main electrode layer **542**, for example, Al containing 1% Cu is used. A protection layer **55** is formed in such a manner as to cover the comb-shaped electrodes **60***a* and **60***b*. The protection layer **55** is a layer for protecting the main electrode layer **542** from an external environment, controlling a frequency temperature characteristic, increasing anti-humidity, and the like and is a dielectric film having, for example, silicon dioxide as a main component.

[0057] The materials of the close-contact layer **540**, the main electrode layer **542**, and the protection layer **55** are not limited to the materials described above. Further, the IDT electrode **54** does not have to have the layered structure above. The IDT electrode **54** may be formed from a metal, such as Ti, Al, Cu, Pt, Au, Ag, or Pd, or an alloy thereof and may also be composed of a plurality of multilayer bodies formed from the metal or the alloy above. The protection layer **55** does not have to be formed.

[0058] The layered structure of the piezoelectric substrate **50** will then be described.

[0059] As illustrated in FIG. **2**AC, the piezoelectric substrate **50** includes a high-acoustic-velocity supporting substrate **51**, a low-acoustic-velocity film **52**, and a piezoelectric film **53** and has a structure in which the high-acoustic-velocity supporting substrate **51**, the low-acoustic-velocity film **52**, and the piezoelectric film **53** are stacked in this order.

[0060] The piezoelectric film **53** is formed from, for example, a θ° Y-cut X-propagating $LiTaO_3$ piezoelectric single crystal or a piezoelectric ceramic (a lithium tantalite single crystal or a ceramic that is cut along a plane with a normal line serving as an axis rotated by 8° from a Y axis with respect to an X axis serving as the center axis and in which a surface acoustic wave propagates in the X-axis direction). The material and the cut angle θ of the piezoelectric single crystal used as the piezoelectric film **53** are appropriately selected in accordance with the required specifications of the filters.

[0061] The high-acoustic-velocity supporting substrate **51** is a substrate that supports the low-acoustic-velocity film **52**, the piezoelectric film **53**, and the IDT electrode **54**. Further, the high-acoustic-velocity supporting substrate **51** is a substrate in which the acoustic velocity of a bulk wave in the high-acoustic-velocity supporting substrate **51** is higher than that of an acoustic wave such as the surface acoustic wave propagating in the piezoelectric film **53** or a boundary acoustic wave. The high-acoustic-velocity supporting substrate **51** confines the surface acoustic wave in a portion where the piezoelectric film **53** and the low-acoustic-velocity film **52** are laminated and functions to prevent the surface acoustic wave from leaking to a portion lower than the high-acoustic-velocity supporting substrate **51**. The high-acoustic-velocity supporting substrate **51** is, for example, a silicon substrate. As the material of the high-acoustic-velocity supporting substrate **51**, for example, a piezoelectric body, such as aluminum nitride, lithium tantalate, lithium niobate, or crystal; ceramic, such as alumina, sapphire, magnesia, silicon nitride, silicon carbide, zirconia, cordierite, mullite, steatite, forsterite, spinel, or sialon; a dielectric, such as aluminum oxide, silicon oxynitride, diamond-like carbon (DLC), or diamond; a semiconductor such as silicon; or a material having any of the materials described above serving as a main component may be used. Spinel described above includes an aluminum compound containing one or more chemical elements selected from Mg, Fe, Zn, and Mn, and other elements and oxygen. As examples of spinel described above, $MgAl_2O_4$, $FeAl_2O_4$, $ZnAl_2O_4$, and $MnAl_2O_4$ may be cited.

[0062] The low-acoustic-velocity film **52** is a film in which the acoustic velocity of a bulk wave in the low-acoustic-velocity film **52** is lower than that of a bulk wave propagating in the piezoelectric film **53** and is disposed between the piezoelectric film **53** and the high-acoustic-velocity supporting substrate **51**. This structure and the nature of concentration of the energy of the acoustic wave in a medium naturally having a low acoustic velocity prevent the surface acoustic wave energy from leaking to the outside of the piezoelectric film **53**. As the material of the low-acoustic-velocity film **52**, for example, a dielectric such as a compound obtained by adding fluorine, carbon, or boron to glass, silicon oxide, silicon oxynitride, lithium oxide, tantalic acid anhydride, or silicon oxide; or a material having any of the materials described above serving as a main component may be used.

[0063] The layered structure described above of the piezoelectric substrate **50** enables a Q value in the resonant frequency and the anti-resonant frequency to be made considerably higher than that in a structure in the related art in which a single layer piezoelectric substrate is used. That is, an acoustic wave resonator having a high Q value may be formed, and thus a filter having a low insertion loss may be formed by using the acoustic wave resonator.

[0064] The high-acoustic-velocity supporting substrate **51** may have a structure in which the supporting substrate and a high-acoustic-velocity film are laminated, the high-acoustic-velocity film having a higher acoustic velocity of the propagating bulk wave than the acoustic velocity of an acoustic wave such as a surface acoustic wave propagating in the piezoelectric film **53** or a boundary acoustic wave. In this case, as the material of the high-acoustic-velocity film, the same material as the material of the high-acoustic-velocity supporting substrate **51** may be used. As the material of the supporting substrate, for example, a piezoelectric body, such as aluminum nitride, lithium tantalate, lithium niobate, or crystal; ceramic, such as alumina, sapphire, magnesia, silicon nitride, silicon carbide, zirconia, cordierite, mullite, steatite, or forsterite; dielectric such as diamond or glass; a semiconductor, such as silicon or gallium nitride; a resin; or a material having any of the materials described above serving as a main component may be used.

[0065] In this specification, the phrase "the main component of a material" refers to a component in which the percentage of the component in the material exceeds 50% by weight. The main component described above may exist in one of single crystal, polycrystalline, and amorphous states or in a mixed state.

[0066] FIG. **2**B is a cross-sectional view schematically illustrating a second example of the acoustic wave resonator included in the acoustic wave filter **1** according to the embodiment. For the surface acoustic wave resonator **60** illustrated in FIGS. **2**AA-**2**AC, the example where the IDT electrode **54** is formed on the piezoelectric substrate **50** having the piezoelectric film **53** has been described; however, as illustrated in FIG. **2**B, the substrate where the IDT electrode **54** is formed may be a single crystal piezoelectric substrate **57** composed of a single layer as a piezoelectric layer.

[0067] The single crystal piezoelectric substrate **57** is formed from, for example, a $LiNbO_3$ piezoelectric single crystal. The acoustic wave resonator according to this example includes the $LiNbO_3$ single crystal piezoelectric substrate **57**, the IDT electrode **54**, and a protection layer **58** formed on the single crystal piezoelectric substrate **57** and the IDT electrode **54**.

[0068] The layered structure, the material, the cut angles, and the thickness of the piezoelectric film **53** and the single crystal piezoelectric substrate **57** that are described above may be appropriately changed depending on the required bandpass characteristics or the like of an acoustic wave filter device. Even an acoustic wave resonator using a $LiTaO_3$ piezoelectric substrate or the like having cut angles other than the cut angles described above may exert the same effects as those of the surface acoustic wave resonator **60** using the piezoelectric film **53** described above.

[0069] The substrate where the IDT electrode **54** is formed may have a structure in which a supporting substrate, an energy confinement layer, and a piezoelectric film are laminated in this order. The IDT electrode **54** is formed on the piezoelectric film. For the piezoelectric film, for example, a $LiTaO_3$ piezoelectric single crystal or a piezoelectric ceramic is used. The supporting substrate is a substrate that supports the piezoelectric film, the energy confinement layer, and the IDT electrode **54**.

[0070] The energy confinement layer is composed of one or more layers, and the velocity of a bulk acoustic wave propagating in at least one of the layers is higher than the velocity of an acoustic wave propagating near the piezoelectric film. For example, the energy confinement layer may have a layered structure having a low-acoustic-velocity layer and a high-acoustic-velocity layer. The low-acoustic-velocity layer is a film in which the acoustic velocity of the acoustic wave propagating in the piezoelectric film is lower than the acoustic velocity of a bulk wave in the low-acoustic-velocity layer. The high-acoustic-velocity layer is a film in which the acoustic velocity of a bulk wave in the high-acoustic-velocity layer is higher than the acoustic velocity of the acoustic wave propagating in the piezoelectric film. The supporting substrate may be used as the high-acoustic-velocity layer.

[0071] The energy confinement layer may also be an acoustic impedance layer having a structure in which a low-acoustic-impedance layer having lower acoustic impedance and a high-acoustic-impedance layer having higher acoustic impedance are alternately stacked.

[0072] Electrode parameters of the IDT electrode **54** included in the surface acoustic wave resonator **60** is described.

[0073] The wavelength of an acoustic wave resonator is defined by the wavelength λ that is the repetition period of the electrode fingers **61***a* or **61***b* included in the IDT electrode **54** illustrated in FIG. **2**AB. An electrode finger pitch is ½ of the wavelength λ. In a case where the line width of the electrode fingers **61***a* and **61***b* included in each of the comb-shaped electrodes **60***a* and **60***b* is W, and the width of a space between the electrode finger **61***a* and the electrode finger **61***b* adjacent to each other is S, the electrode finger pitch is defined as (W+S). The duty of the IDT electrode **54** is the line width share of each of the electrode fingers **61***a* and **61***b*, thus is a ratio of the line width of the electrode fingers **61***a* and **61***b* to a value obtained by adding the line width to the space width, and thus is defined as W/(W+S). The interdigital width of the IDT electrode **54** is a length of the electrode fingers overlapping when the electrode fingers **61***a* and the electrode fingers **61***b* are seen in the acoustic wave propagation direction (X-axis direction).

[0074] If a distance between adjacent electrode fingers is not fixed in the IDT electrode **54**, the electrode finger pitch of the IDT electrode **54** is defined by the average electrode finger pitch of the IDT electrode **54**. In a case where the total number of electrode fingers of the electrode fingers **61***a* and **61***b* included in the IDT electrode **54** is Ni, a distance between respective centers of the electrode finger located on one end, of the IDT electrode **54**, in the acoustic wave propagation direction and the electrode finger located on the other end is Di, the average electrode finger pitch of the IDT electrode **54** is defined as Di/(Ni−1).

[0075] FIG. **2**C is a cross-sectional view schematically illustrating a third example of the acoustic wave resonator included in the acoustic wave filter **1** according to the embodiment. FIG. **2**C illustrates a bulk acoustic wave resonator as an acoustic wave resonator of the acoustic wave filter **1**. As illustrated in FIG. **2**C, the bulk acoustic wave resonator has, for example, a supporting substrate **65**, a lower electrode **66**, a piezoelectric layer **67**, and an upper electrode **68** and has a configuration in which the supporting substrate **65**, the lower electrode **66**, the piezoelectric layer **67**, and the upper electrode **68** are stacked in this order.

[0076] The supporting substrate **65** is a substrate for supporting the lower electrode **66**, the piezoelectric layer **67**, and the upper electrode **68** and is, for example, a silicon substrate. The supporting substrate **65** has a hollow portion in a part of area in contact with the lower electrode **66**. This enables the piezoelectric layer **67** to vibrate freely.

[0077] The lower electrode **66** is an example of a first planar electrode and is formed on one of the surfaces of the supporting substrate **65**. The upper electrode **68** is an example of a second planar electrode and is formed on the one surface of the supporting substrate **65**. As the material of the lower electrode **66** and the upper electrode **68**, for example, Al containing 1% Cu is used.

[0078] The piezoelectric layer **67** is an example of a piezoelectric thin film and is formed between the lower electrode **66** and the upper electrode **68**. The piezoelectric layer **67** contains as a main component, at least one of, for example, zinc oxide (ZnO), aluminum nitride (AlN), lead zirconate titanate (PZT), potassium niobate (KN), lithium niobate (LN), lithium tantalate (LT), crystal, and lithium borate (LiBO).

[0079] The bulk acoustic wave resonator having the layered structure described above induces a bulk acoustic wave in the piezoelectric layer **67** by applying electrical energy between the lower electrode **66** and the upper electrode **68** and thus generates resonance. The bulk acoustic wave generated by this bulk acoustic wave resonator propagates in a portion between the lower electrode **66** and the upper electrode **68** in a direction perpendicular to the film surface of the piezoelectric layer **67**. The bulk acoustic wave resonator is thus a resonator using the bulk acoustic wave.

3 Resonance Characteristics and Bandpass Characteristics of Acoustic Wave Filter **1**

[0080] First, the basic operating principle of a ladder bandpass filter composed of one serial arm resonator and one parallel arm resonator is described.

[0081] The parallel arm resonator has a resonant frequency frp and an anti-resonant frequency fap (>frp), and the serial arm resonator has a resonant frequency frs and an anti-resonant frequency fas (>frs>frp). In the serial arm resonator and the parallel arm resonator having the resonance characteristic, the anti-resonant frequency fap of the parallel arm resonator and the resonant frequency frs of the serial arm resonator are typically made close to each other. This causes a frequency near the resonant frequency frp where the impedance of the parallel arm resonator is close to 0 to fall under a stopband on the low frequency side. If the frequency is increased further, the impedance of the parallel arm resonator is high near the anti-resonant frequency fap, and the impedance of the serial arm resonator is close to 0 near the resonant frequency frs. This causes a frequency near the anti-resonant frequency fap to the resonant frequency frs to fall under a signal pass band in a signal path serving as a serial arm path. A pass band on which the electrode parameter and the electromechanical coupling coefficient of the acoustic wave resonator are reflected may thereby be formed. Further, when the frequency is high and close to the anti-resonant frequency fas, the impedance of the serial arm resonator is high, and the frequency falls under a stopband on the high frequency side.

[0082] In each of the serial arm resonator and the parallel arm resonator, the impedance of the resonator represents capacitive impedance (capacitance) in a frequency band lower than the resonant frequency and inductive impedance (inductance) in a frequency band higher than the resonant frequency and lower than the anti-resonant frequency. The impedance of the resonator represents capacitive impedance in a frequency band higher than the anti-resonant frequency.

[0083] The impedance characteristics and bandpass characteristics of the acoustic wave filter **1** according to this embodiment and acoustic wave filters according to comparative examples will then be described.

[0084] FIG. **3**A is a view of the circuit configuration of an acoustic wave filter **200** according to Comparative Example 1. FIGS. **3**BA and **3**BB illustrate a Smith chart representing the bandpass characteristic and the impedance of the acoustic wave filter **200** according to Comparative Example 1. As illustrated in FIG. **3**A, the acoustic wave filter **200** according to Comparative Example 1 has a configuration in which the serial arm resonator **13** and the inductor **31** are not disposed, as compared with the acoustic wave filter **1** according to the embodiment. The acoustic wave filter **200** according to Comparative Example 1 is thus a ladder acoustic wave filter including the serial arm resonators **11** and **12** and the parallel arm resonator **21**.

[0085] In the following description, the acoustic wave filter **1** according to this embodiment and the acoustic wave filter according to Comparative Example have a pass band including, for example, Band B**40**A for LTE or Band n**40**A (2300 to 2370 MHz) for 5G-NR.

[0086] In the acoustic wave filter **200**, the resonant frequencies of the serial arm resonators **11** and **12** and the anti-resonant frequency of the parallel arm resonator **21** are located in the pass band of the acoustic wave filter **200** based on basic operating principle of the ladder bandpass filter, and thereby the bandpass characteristic of the bandpass filter is obtained, as illustrated in FIG. **3**BA. The impedance of the pass band is located on the capacitive and high impedance side. Accordingly, in a case where the low-noise amplifier **2** having the capacitive input impedance is connected to the input/output terminal **120**, a matching element having inductive impedance and low transmission loss is required between the acoustic wave filter **200** and the low-noise amplifier **2**.

[0087] FIGS. **4**A-**4**C illustrate a Smith chart representing the bandpass characteristic of the acoustic wave filter **1** according to the embodiment, the capacitance characteristic of the serial arm resonator **13**, and the impedance of the acoustic wave filter **1**. As compared with the acoustic wave filter **200** according to Comparative Example 1, the acoustic wave filter **1** according to the embodiment additionally has the serial arm resonator **13** and the inductor **31**.

[0088] As illustrated in FIG. **4**A, the serial arm resonator **13** has a resonant frequency frs**13** leading to the minimum impedance and an anti-resonant frequency fas**13** leading to the maximum impedance. The resonant frequency frs**13** is located at the highest frequency end of the pass band of the acoustic wave filter **1**. This causes the impedance of the serial arm resonator **13** in the pass band and the DC region of the acoustic wave filter **1** to be capacitive impedance. The capacitance value of the serial arm resonator **13** in the DC region is approximately 5 pF and is a sufficiently high capacitance value to block a DC component. Accordingly, even if a DC blocking capacitor is not disposed in series to a path connecting the acoustic wave filter **1** and the input port of the amplifying transistor of the low-noise amplifier **2**, DC bias current to be supplied to the low-noise amplifier **2** may be prevented from leaking to the ground via the inductor **31**.

[0089] In addition, as illustrated in FIG. **4**B, the capacitance value of the serial arm resonator **13** at the highest frequency end of the pass band of the acoustic wave filter **1** increases in an exponential function manner and reaches 100 pF or higher, and the capacitance value of the serial arm resonator **13** at the lowest frequency end of the pass band also increases as compared with the capacitance value in the DC region and is approximately 13 pF. That is, the capacitance value of the serial arm resonator **13** in the pass band is an exceedingly high value as compared with the capacitance value in the DC region, impedance determined from the inverse of the capacitance value is low, and the serial arm resonator **13** is substantially in a short-circuited state.

[0090] Since the serial arm resonator **13** is substantially in the short-circuited state in the frequency band as the pass band described above, the inductor **31** thereby functions as the matching element having inductive impedance without necessarily being influenced by the serial arm resonator **13**. Accordingly, if an external circuit is connected to the input/output terminal **120** having the capacitive impedance, matching loss in the pass band may be reduced, and the acoustic wave filter **1** in which the DC blocking in the DC region and the impedance matching in the pass band are ensured may be provided. In addition, DC bias current may be supplied to the low-noise amplifier **2** connected to the input/output terminal **120** with high accuracy, and matching loss of the radio frequency reception signal may be reduced. The radio frequency module **100** with reduced deterioration of the noise figure of the low-noise amplifier **2** may be provided.

[0091] The resonant frequency frs**13** of the serial arm resonator **13** may be located at a frequency higher than or equal to the highest frequency end of the pass band. This enables the impedance of the serial arm resonator **13** in the pass band and the DC region to be capacitive impedance and enables the impedance in the pass band to be higher than the impedance in the DC region. The DC blocking and the impedance matching in the pass band may thus be ensured.

[0092] In addition, in this embodiment, the serial arm resonator **13** has the highest electrostatic capacitance among the serial arm resonators **11** to **13** and the parallel arm resonator **21**.

[0093] This enables the capacitance value of the serial arm resonator **13** in the DC region to be the highest among the acoustic wave resonators included in the acoustic wave filter **1** and the impedance of the serial arm resonator **13** in the pass band to be exceedingly low. Accordingly, the matching loss in the pass band may be reduced considerably, and the acoustic wave filter **1** in which the DC blocking in the DC region and the impedance matching in the pass band are ensured may be provided.

[0094] The serial arm resonator **13** does not have to have the resonant frequency frs**13** located at the frequency higher than or equal to the highest frequency end of the pass band and may have the highest resonant frequency among the serial arm resonators **11** to **13**. This enables the impedance of the serial arm resonator **13** in the DC region to be capacitive impedance, the impedance of the serial arm resonator **13** in the pass band to be substantially capacitive impedance, and the impedance in the pass band to be higher than the impedance in the DC region. The DC blocking and the impedance matching in the pass band may thus be ensured. Further, since the resonant frequency frs**13** of the serial arm resonator **13** is located in a higher frequency region of the pass band, insertion loss in the higher frequency region may be reduced, and what is called a drop of the bandpass characteristic in the highest frequency end in the pass band described above may be prevented.

[0095] FIGS. **5**A-**5**C illustrate a Smith chart representing the bandpass characteristic of an acoustic wave filter **500** according to Comparative Example 2, the capacitance characteristic of a serial arm resonator **513**, and the impedance of the acoustic wave filter **500**. The acoustic wave filter **500** according to Comparative Example 2 includes the serial arm resonators **11**, **12**, and **513**, the parallel arm resonator **21**, the inductor **31**, and the input/output terminals **110** and **120**. The acoustic wave filter **500** according to Comparative Example 2 is different from the acoustic wave filter **1** according to the embodiment only in that the serial arm resonator **13** is replaced with the serial arm resonator **513**. Hereinafter, the acoustic wave filter **500** according to Comparative Example 2 will be described in such a manner that the description of the same configuration as that of the acoustic wave filter **1** according to the embodiment is omitted and focus is placed on a different configuration.

[0096] The serial arm resonators **11**, **12**, and **513** are connected in the order of the serial arm resonators **11**, **12**, and **513** from the input/output terminal **110**. The serial arm resonator **513** is connected closest to the input/output terminal **120** among the serial arm resonators **11**, **12**, and **513**, the parallel arm resonator **21**, and the inductor **31**.

[0097] The serial arm resonator **513** has the highest electrostatic capacitance among the serial arm resonators **11**, **12**, and **513** and the parallel arm resonator **21**. In addition, as illustrated in FIG. **5**A, a resonant frequency frs**513** of the serial arm resonator **513** is located near the center of the pass band of the acoustic wave filter **500**.

[0098] The serial arm resonator **513** has the highest electrostatic capacitance among the serial arm resonators **11**, **12**, and **513** and the parallel arm resonator **21**. In addition, as illustrated in FIG. **5**A, the impedance of the serial arm resonator **513** in the DC region is capacitive impedance. The capacitance value of the serial arm resonator **513** in the DC region is approximately 5 pF and is a sufficiently high capacitance value to block the DC component. Accordingly, DC bias current to be supplied to the low-noise amplifier **2** may be prevented from leaking to the ground via the inductor **31**.

[0099] In contrast, as illustrated in FIGS. **5**A and **5**B, the serial arm resonator **513** has capacitive impedance at lower frequencies in the pass band and has a high capacitance value; however, the serial arm resonator **513** has inductive impedance at higher frequencies in the pass band and has a low capacitance value. This causes the serial arm resonator **513** to be in the short-circuited state at the lower frequencies in the pass band but not to be in the short-circuited state at the higher frequencies in the pass band. The inductor **31** is thereby influenced by the serial arm resonator **13** at the higher frequencies in the pass band, and thus the degree of the impedance matching is lowered.

[0100] Further, as illustrated in FIG. **5**C, in the acoustic wave filter **500** according to Comparative Example 2, the serial arm resonator **513** has capacitive impedance at lower frequencies in the pass band, and a constant resistance circle is shifted counterclockwise. The serial arm resonator **513** has inductive impedance at higher frequencies in the pass band, and the constant resistance circle is shifted clockwise. This causes the spread (R**500** in FIG. **5**C) of the impedance trajectory in the pass band of the acoustic wave filter **500** to be larger than the spread (R**1** in FIG. **4**C) of the impedance trajectory in the pass band of the acoustic wave filter **1**. That is, the degree of impedance concentration in the pass band of the acoustic wave filter **500** is lowered as compared with the degree of impedance concentration in the pass band of the acoustic wave filter **1**. Accordingly, signal transmission loss in the pass band of the acoustic wave filter **500** is increased.

[0101] FIGS. **6**A-**6**C illustrates a Smith chart representing the bandpass characteristic of an acoustic wave filter **600** according to Comparative Example 3, the capacitance characteristic of a serial arm resonator **613**, and the impedance of the acoustic wave filter **600**. The acoustic wave filter **600** according to Comparative Example 3 includes the serial arm resonators **11**, **12**, and **613**, the parallel arm resonator **21**, the inductor **31**, and the input/output terminals **110** and **120**. The acoustic wave filter **600** according to Comparative Example 3 is different from the acoustic wave filter **1** according to the embodiment only in that the serial arm resonator **13** is replaced with the serial arm resonator **613**. Hereinafter, the acoustic wave filter **600** according to Comparative Example 3 will be described in such a manner that the description of the same configuration as that of the acoustic wave filter **1** according to the embodiment is omitted and focus is placed on a different configuration.

[0102] The serial arm resonators **11**, **12**, and **613** are connected in the order of the serial arm resonators **11**, **12**, and **613** from the input/output terminal **110**. The serial arm resonator **613** is connected closest to the input/output terminal **120** among the serial arm resonators **11**, **12**, and **613**, the parallel arm resonator **21**, and the inductor **31**.

[0103] The serial arm resonator **613** has the highest electrostatic capacitance among the serial arm resonators **11**, **12**, and **613** and the parallel arm resonator **21**. In addition, as illustrated in FIG. **6**A, a resonant frequency frs**613** and an anti-resonant frequency fas**613** of the serial arm resonator **613** are located at frequencies lower than the lowest frequency end of the pass band of the acoustic wave filter **600**.

[0104] The serial arm resonator **613** has the highest electrostatic capacitance among the serial arm resonators **11**, **12**, and **613** and the parallel arm resonator **21**. In addition, as illustrated in FIG. **6**A, the impedance of the serial arm resonator **613** in the DC region is capacitive impedance. The capacitance value of the serial arm resonator **613** in the DC region is higher than or equal to 5 pF and is a sufficiently high capacitance value to block the DC component. Accordingly, DC bias current to be supplied to the low-noise amplifier **2** may be prevented from leaking to the ground via the inductor **31**.

[0105] In contrast, as illustrated in FIGS. **6**A and **6**B, the impedance of the serial arm resonator **613** in the pass band of the acoustic wave filter **600** is capacitive impedance but has a low capacitance value. Accordingly, the impedance of the serial arm resonator **613** in the pass band is high. As illustrated in FIG. **6**C, even if the inductor **31** causes the impedance of the acoustic wave filter **600** in the pass band to move to an inductive region P**1**, the serial arm resonator **613** having the high capacitive impedance causes the impedance of the acoustic wave filter **600** in the pass band to move largely counterclockwise in the constant resistance circle and thus leads to low inductive reactance. Impedance matching with the low-noise amplifier **2** connected to the input/output terminal **120** is thereby insufficient, and the matching loss of the radio frequency reception signal is increased.

4 Terminal Layout of Acoustic Wave Filter **1**

[0106] The terminal layout of the acoustic wave filter **1** according to this embodiment will then be described. FIG. **7**A is a view of the circuit configuration of the acoustic wave filter **1** according to the embodiment. FIG. **7**B is a plan view of the electrode layout of the acoustic wave filter **1** according to the embodiment.

[0107] As illustrated in FIG. **7**A, the acoustic wave filter **1** includes a substrate **70** in addition to the circuit configuration of the acoustic wave filter **1** illustrated in FIG. **1**. The serial arm resonators **11** to **13**, the parallel arm resonator **21**, the input/output terminals **110** and **120**, a terminal **150**, and a ground terminal **160** are disposed on the substrate **70**. This enables the acoustic wave filter **1** to be downsized.

[0108] The terminal **150** is an example of a first terminal, is a node on a path connecting the serial arm resonators **12** and **13**, and is connected to the inductor **31**. The ground terminal **160** is connected to the parallel arm resonator **21** and is set at the ground potential.

[0109] The substrate **70** is an example of a first substrate and has main surfaces **70***a* and **70***b* facing each other. If each of the serial arm resonators **11** to **13** and the parallel arm resonator **21** is the surface acoustic wave resonator including the IDT electrode **54**, the substrate **70** has piezoelectricity and corresponds to the piezoelectric substrate **50** illustrated in FIGS. **2**AA-**2**AC or the single crystal piezoelectric substrate **57** illustrated in FIG. **2**B. If each of the serial arm resonators **11** to **13** and the parallel arm resonator **21** is a bulk acoustic wave resonator including a multilayer body having the lower electrode **66**, the piezoelectric layer **67**, and the upper electrode **68** in this order from the main surface **70***a* or **70***b*, the substrate **70** includes silicon and corresponds to the supporting substrate **65** illustrated in FIG. **2**C.

[0110] As illustrated in FIG. **7**B, the input/output terminals **110** and **120**, the terminal **150**, and the ground terminal **160** are disposed on the main surface **70***a*. The IDT electrode and the multilayer body described above may be disposed on any of the main surfaces **70***a* and **70***b*. In plan view of the main surface **70***a*, the ground terminal **160** is disposed between the terminal **150** and the input/output terminal **110**.

[0111] Since the ground terminal **160** is disposed between the terminal **150** and the input/output terminal **110**, deterioration of the attenuation band of the acoustic wave filter **1** due to the electromagnetic coupling (electrical field coupling or magnetic coupling) between the inductor **31** connected to the terminal **150** and the input/output terminal **110** may thereby be prevented.

[0112] The ground terminal disposed between the terminal **150** and the input/output terminal **110** does not have to be the ground terminal **160** and may be a different ground terminal (GND).

[0113] FIG. **7**C is a plan view of the electrode layout of an acoustic wave filter **1**A according to Modification 1 of the embodiment. The acoustic wave filter **1**A according to Modification 1 has the same circuit configuration as that of the acoustic wave filter **1** according to the embodiment and is different only in the terminal configuration. Accordingly, the terminal configuration of the acoustic wave filter **1**A according to Modification 1 will hereinafter be described.

[0114] As illustrated in FIG. **7**C, a plurality of ground terminals (GNDs) are disposed on the main surface **70***a*. In plan view of the main surface **70***a*, a virtual line LL (first virtual line) connecting the terminal **150** and the input/output terminal **110** crosses a virtual line LG (second virtual line) connecting two of the plurality of ground terminals (GNDs), and a length DG of the virtual line LG is shorter than a length DL of the virtual line LL.

[0115] Since the electromagnetic shielding region composed of the plurality of ground terminals (GNDs) is formed between the terminal **150** and the input/output terminal **110**, deterioration of the attenuation band of the acoustic wave filter **1**A due to electromagnetic coupling between the inductor **31** connected to the terminal **150** and the input/output terminal **110** may thereby be prevented.

[0116] The ground terminal disposed to cross the virtual line connecting the terminal **150** and the input/output terminal **110** does not have to include the ground terminal **160** and may be a different ground terminal (GND).

5 Component Layout of Radio Frequency Module **100**A

[0117] The circuit configuration and the component configuration of a radio frequency module **100**A according to Modification 2 of this embodiment will then be described.

[0118] FIG. **8**A is a view of the circuit configuration of the radio frequency module **100**A according to Modification 2 of the embodiment. FIG. **8**B is a plan view of the component configuration of the radio frequency module **100**A according to Modification 2 of the embodiment.

[0119] As illustrated in FIG. **8**A, the radio frequency module **100**A according to this modification includes the acoustic wave filter **1**, the low-noise amplifier **2**, a filter **3**, a switch **40**, and the inductor **32**.

[0120] The acoustic wave filter **1** is the acoustic wave filter **1** according to the embodiment and is connected to the input terminal **130** of the low-noise amplifier **2** with the switch **40** and the inductor **32** interposed therebetween. The filter **3** is an example of a first filter and has one end connected to the input terminal **130** with the switch **40** and the inductor **32** interposed therebetween and the other end connected to an input/output terminal **170**.

[0121] The switch **40** has a common terminal **40***a*, a selection terminal **40***b* (first selection terminal), and a selection terminal **40***c* (second selection terminal) and performs switching between connection between the common terminal **40***a* and the selection terminal **40***b* and connection between the common terminal **40***a* and the selection terminal **40***c*. The common terminal **40***a* is connected to the input terminal **130** with the inductor **32** interposed therebetween, the selection terminal **40***b* is connected to the input/output terminal **120**, and the selection terminal **40***c* is connected to the filter **3**.

[0122] The configuration above enables the radio frequency module **100**A to select one of a reception signal having passed through the acoustic wave filter **1** and a reception signal having passed through the filter **3** and to amplify the reception signal at the low-noise amplifier **2**.

[0123] In the radio frequency module **100**A, at least one of the switch **40** and the inductor **32** does not have to be provided.

[0124] As illustrated in FIG. **8**B, the radio frequency module **100**A further includes a mounting substrate **90**. On the mounting substrate **90**, the acoustic wave filter **1**, the filter **3**, and the low-noise amplifier **2** are disposed. The switch **40** and the inductor **32** may be disposed on the mounting substrate **90**.

[0125] As the mounting substrate **90**, for example, one of a low temperature co-fired ceramics (LTCC) substrate and a high temperature co-fired ceramics (HTCC) substrate that have a layered structure of a plurality of dielectric layers, a substrate having components built therein, a substrate having a redistribution layer (RDL), a printing circuit board, or the like may be used, but the mounting substrate **90** is not limited to these.

[0126] A distance D**12** between the low-noise amplifier **2** and the acoustic wave filter **1** is longer than a distance D**32** between the low-noise amplifier **2** and the filter **3**.

[0127] Even if the wiring element connecting the acoustic wave filter **1** and the low-noise amplifier **2** is long, the inductor **31** included in the acoustic wave filter **1** may thereby override the influence of parasitic capacitance occurring on the wiring element. The filter **3** desirably having a short wiring element for connection to the low-noise amplifier **2** may thereby be disposed between the acoustic wave filter **1** and the low-noise amplifier **2**, and thus the radio frequency module **100**A that is downsized and has reduced signal transmission loss may be provided.

6 Effects and the Like

[0128] As described above, the acoustic wave filter **1** according to this embodiment includes the plurality of serial arm resonant devices including the serial arm resonators **12** and **13** disposed on the serial arm path connecting the input/output terminals **110** and **120**, the one or more parallel arm resonant devices including the parallel arm resonator **21** connected between the serial arm path and the ground, and the inductor **31** connected between the ground and the first path connecting the serial arm resonators **12** and **13**. The serial arm resonator **13** is connected closest to the input/output terminal **120** among the plurality of serial arm resonant devices, the one or more parallel arm resonant devices, and the inductor **31**, and the serial arm resonator **13** has the highest resonant frequency among the plurality of serial arm resonant devices.

[0129] Since the serial arm resonator **13** has the highest the resonant frequency, the impedance of the serial arm resonator **13** in the pass band the DC region of the acoustic wave filter **1** is thereby capacitive impedance. Accordingly, even if a DC blocking capacitor is not disposed in series to the path connecting the input/output terminal **120** and the external circuit, a DC component may be prevented from leaking to the ground via the inductor **31**. In addition, the capacitance value of the serial arm resonator **13** in the pass band is an exceedingly high value as compared with the capacitance value in the DC region, the impedance of the serial arm resonator **13** in the pass band is low, and the serial arm resonator **13** is substantially in the short-circuited state. The inductor **31** thereby functions as the matching element having inductive impedance in the frequency band as the pass band without necessarily being influenced by the serial arm resonator **13**. Accordingly, if the external circuit is connected to the input/output terminal **120** having the capacitive impedance, the matching loss in the pass band may be reduced, and the acoustic wave filter **1** in which the DC blocking and the impedance matching in the pass band are ensured may be provided. In addition, in the radio frequency module **100** having the low-noise amplifier **2** serving as the external circuit connected to the input/output terminal **120**, DC bias current may be supplied to the low-noise amplifier **2** with high accuracy, and the matching loss of the radio frequency reception signal may be reduced. Accordingly, the noise figure of the low-noise amplifier **2** may be prevented from being deteriorated.

[0130] For example, in the acoustic wave filter **1**, the serial arm resonator **13** has the highest electrostatic capacitance among the plurality of serial arm resonant devices and the one or more parallel arm resonant devices.

[0131] This enables the capacitance value of the serial arm resonator **13** in the DC region to be the highest among the acoustic wave resonators included in the acoustic wave filter **1** and the impedance of the serial arm resonator **13** in the pass band to be exceedingly low. Accordingly, the matching loss in the pass band may be reduced considerably, and the acoustic wave filter **1** in which the DC blocking in the DC region and the impedance matching in the pass band are ensured may be provided.

[0132] For example, in the acoustic wave filter **1**, the resonant frequency frs**13** of the serial arm resonator **13** is located at a frequency higher than or equal to the highest frequency end of the pass band of the acoustic wave filter **1**.

[0133] This causes the entire pass band of in the acoustic wave filter **1**, the impedance of the serial arm resonator **13** to be capacitive impedance, and the matching loss in the pass band may be reduced considerably if the external circuit having capacitive impedance is connected to the input/output terminal **120**.

[0134] For example, the acoustic wave filter **1** further includes the substrate **70** having the main surfaces **70***a* and **70***b* facing each other. Each acoustic wave resonator included in a corresponding one of the serial arm resonators **11** to **13** and the parallel arm resonator **21** is formed on the substrate **70**, and the input/output terminals **110** and **120**, the terminal **150** on the first path connected to the inductor **31**, and the ground terminal **160** are disposed on the main surface **70***a*.

[0135] This causes the acoustic wave resonators included in the acoustic wave filter **1** to be integrated on the substrate **70**, and thus the acoustic wave filter **1** may be downsized.

[0136] For example, in the acoustic wave filter **1**, in plan view of the main surface **70***a*, the ground terminal **160** is disposed between the terminal **150** and the input/output terminal **110**.

[0137] Deterioration of the attenuation band of the acoustic wave filter **1** due to electromagnetic coupling between the inductor **31** connected to the terminal **150** and the input/output terminal **110** may thereby be prevented.

[0138] For example, in the acoustic wave filter **1**A according to Modification 1, the plurality of ground terminals are disposed on the main surface **70***a*. In addition, in plan view of the main surface **70***a*, the virtual line LL connecting the terminal **150** and the input/output terminal **110** crosses the virtual line LG connecting two of the plurality of ground terminals, and the virtual line LG is shorter than the virtual line LL.

[0139] Since the electromagnetic shielding region composed of the plurality of ground terminals is formed between the terminal **150** and the input/output terminal **110**, deterioration of the attenuation band of the acoustic wave filter **1**A due to electromagnetic coupling between the inductor **31** connected to the terminal **150** and the input/output terminal **110** may thereby be prevented.

[0140] For example, in the acoustic wave filter **1** (**1**A), the substrate **70** has piezoelectricity, and each acoustic wave resonator included in a corresponding one of the serial arm resonators **11** to **13** and the parallel arm resonator **21** includes the IDT electrode.

[0141] The acoustic wave filter **1** (**1**A) is thereby the ladder filter composed of the surface acoustic wave resonators.

[0142] For example, in the acoustic wave filter **1** (**1**A), the substrate **70** includes silicon, each acoustic wave resonator included in a corresponding one of the serial arm resonators **11** to **13** and the parallel arm resonator **21** includes the multilayer body having the lower electrode **66**, the piezoelectric layer **67**, and the upper electrode **68** in this order from the main surface **70***a* or **70***b*.

[0143] The acoustic wave filter **1** (**1**A) is thereby the ladder filter composed of the bulk acoustic wave resonator.

[0144] The radio frequency module **100** according to this embodiment includes the acoustic wave filter **1** and the low-noise amplifier **2** having the input terminal **130** connected to the input/output terminal **120**.

[0145] The serial arm resonator **13** thereby has the DC blocking function and is in the short-circuited state in the pass band. Accordingly, DC bias current may be supplied to the low-noise amplifier **2** with high accuracy, and the matching loss of the radio frequency reception signal in the pass band may be reduced. The noise figure of the low-noise amplifier **2** may thus be prevented from being deteriorated.

[0146] For example, in the radio frequency module **100**, the capacitor is not disposed in series to the path connecting the input/output terminal **120** and the input terminal of the amplifying transistor included in the low-noise amplifier **2**.

[0147] This enables the deterioration of the noise figure of the low-noise amplifier due to the occurrence of impedance mismatching in the pass band caused by the capacitor.

[0148] For example, the radio frequency module **100**A according to Modification 2 further includes the filter **3** connected to the input terminal **130**, the acoustic wave filter **1**, and the mounting substrate **90** on which the filter **3** and the low-noise amplifier **2** are disposed, and the distance between the low-noise amplifier **2** and the acoustic wave filter **1** is longer than the distance between the low-noise amplifier **2** and the filter **3**.

[0149] Even if the wiring element connecting the acoustic wave filter **1** and the low-noise amplifier **2** is long, the inductor **31** included in the acoustic wave filter **1** may thereby override the influence of parasitic capacitance occurring on the wiring element. The filter **3** desirably having a short wiring element for connection to the low-noise amplifier **2** may thereby be disposed between the acoustic wave filter **1** and the low-noise amplifier **2**, and thus the radio frequency module **100**A that is downsized and has reduced signal transmission loss may be provided.

[0150] For example, the radio frequency module **100**A further includes the switch **40** that has the common terminal **40***a* and the selection terminals **40***b* and **40***c* and performs switching between the connection between the common terminal **40***a* and the selection terminal **40***b* and the connection between the common terminal **40***a* and the selection terminal **40***c*. The common terminal **40***a* is connected to the input terminal **130**, the selection terminal **40***b* is connected to the input/output terminal **120**, and the selection terminal **40***c* is connected to the filter **3**.

[0151] This enables the radio frequency module **100**A to select one of the reception signal having passed through the acoustic wave filter **1** and the reception signal having passed through the filter **3** and to amplify the reception signal at the low-noise amplifier **2**.

Other Embodiments

[0152] The acoustic wave filter and the radio frequency module according to the present disclosure have heretofore been described by using the embodiment and the modifications; however, the present disclosure is not limited to the embodiment and the modifications that are described above. A modification obtained by applying, to the embodiment and the modifications above, a modification conceived of those skilled in the art without necessarily departing from the spirit of the present disclosure, and various types of equipment having the acoustic wave filter and the radio frequency module according to the present disclosure built therein are also included in the present disclosure.

[0153] For example, in the acoustic wave filter and the radio frequency module according to the above-mentioned embodiment and the modifications, a matching element such as an inductor or a capacitor and a switching circuit may be connected between the components.

[0154] The features of the acoustic wave filter and the radio frequency module described based on the above-mentioned embodiment and the modifications will hereinafter be described.

[0155] <1>

[0156] An acoustic wave filter includes:

[0157] a plurality of serial arm resonant devices including a first serial arm resonant device and a second serial arm resonant device, the first serial arm resonant device being disposed on a serial arm path connecting a first input/output terminal and a second input/output terminal;

[0158] one or more parallel arm resonant devices connected between the serial arm path and ground; and

[0159] a first inductor connected between the ground and a first path connecting the first serial arm resonant device and the second serial arm resonant device.

[0160] The first serial arm resonant device is connected closest to the first input/output terminal among the plurality of serial arm resonant devices, the one or more parallel arm resonant devices, and the first inductor, and

[0161] the first serial arm resonant device has a highest electrostatic capacitance among the plurality of serial arm resonant devices and the one or more parallel arm resonant devices and has a highest resonant frequency among the plurality of serial arm resonant devices.

[0162] <2>

[0163] In the acoustic wave filter according to Claim **1**,

[0164] the first serial arm resonant device has highest electrostatic capacitance among the plurality of serial arm resonant devices and the one or more parallel arm resonant devices.

[0165] <3>

[0166] In the acoustic wave filter according to <1> or <2>,

[0167] a resonant frequency of the first serial arm resonant device is located at a frequency higher than or equal to a highest frequency end of a pass band of the acoustic wave filter.

[0168] <4>

[0169] The acoustic wave filter according to any one of <1> to <3> further includes:

[0170] a first substrate having a first main surface and a second main surface facing each other.

[0171] An acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices is formed on the first substrate, and

[0172] the first input/output terminal, the second input/output terminal, a first terminal, and a ground terminal are disposed on the first main surface, the first terminal being connected to the first inductor and being on the first path.

[0173] <5>

[0174] In the acoustic wave filter according to <4>,

[0175] in plan view of the first main surface, the ground terminal is disposed between the first terminal and the second input/output terminal.

[0176] <6>

[0177] In the acoustic wave filter according to <4>,

[0178] a plurality of the ground terminals are disposed on the first main surface.

[0179] In plan view of the first main surface, a first virtual line connecting the first terminal and the second input/output terminal crosses a second virtual line connecting two ground terminals of the plurality of ground terminals, and

[0180] the second virtual line is shorter than the first virtual line.

[0181] <7>

[0182] In the acoustic wave filter according to any one of <4> to <6>,

[0183] the first substrate has piezoelectricity, and

[0184] the acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices includes an IDT electrode.

[0185] <8>

[0186] In the acoustic wave filter according to any one of <4> to <6>,

[0187] the first substrate includes silicon, and

[0188] the acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices includes a multilayer body having a first planar electrode, a piezoelectric thin film, and a second planar electrode in this order from the first main surface or the second main surface.

[0189] <9>

[0190] A radio frequency module includes:

[0191] the acoustic wave filter according to any one of <1> to <8>; and

[0192] a low-noise amplifier having an input terminal connected to the first input/output terminal.

[0193] <10>

[0194] In the radio frequency module according to <9>, a capacitor is not disposed in series to a path connecting the first input/output terminal and an input port of an amplifying transistor included in the low-noise amplifier.

[0195] <11>

[0196] The radio frequency module according to <9> or <10> further includes:

[0197] a first filter connected to the input terminal; and

[0198] a mounting substrate on which the acoustic wave filter, the first filter, and the low-noise amplifier are disposed.

[0199] A distance between the low-noise amplifier and the acoustic wave filter is longer than a distance between the low-noise amplifier and the first filter.

[0200] <12>

[0201] The radio frequency module according to <11> further includes:

[0202] a switch that has a common terminal, a first selection terminal, and a second selection terminal and that performs switching between connection between the common terminal and the first selection terminal and connection between the common terminal and the second selection terminal.

[0203] The common terminal is connected to the input terminal,

[0204] the first selection terminal is connected to the first input/output terminal, and

[0205] the second selection terminal is connected to the first filter.

[0206] The present disclosure may be widely used for communications equipment such as a mobile phone, as an acoustic wave filter and a radio frequency module with low loss that are applicable to a frequency standard for multi-band.

What is claimed is:

1. An acoustic wave filter comprising:

a plurality of serial arm resonant devices including a first serial arm resonant device and a second serial arm resonant device, the first serial arm resonant device being disposed on a serial arm path connecting a first input/output terminal and a second input/output terminal;

one or more parallel arm resonant devices connected between the serial arm path and ground; and

a first inductor connected between ground and a first path connecting the first serial arm resonant device and the second serial arm resonant device,

wherein the first serial arm resonant device is connected closest to the first input/output terminal among the plurality of serial arm resonant devices, the one or more parallel arm resonant devices, and the first inductor, and

wherein the first serial arm resonant device has a highest resonant frequency among the plurality of serial arm resonant devices.

2. The acoustic wave filter according to claim **1**,

wherein the first serial arm resonant device has a highest electrostatic capacitance among the plurality of serial arm resonant devices and the one or more parallel arm resonant devices.

3. The acoustic wave filter according to claim **1**,

wherein a resonant frequency of the first serial arm resonant device is located at a frequency higher than or equal to a highest frequency end of a pass band of the acoustic wave filter.

4. The acoustic wave filter according to claim **1**, further comprising:

a first substrate having a first main surface and a second main surface facing each other,

wherein an acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices is formed on the first substrate, and

wherein the first input/output terminal, the second input/output terminal, a first terminal, and a ground terminal are disposed on the first main surface, the first terminal being connected to the first inductor and being disposed on the first path.

5. The acoustic wave filter according to claim **4**,

wherein in plan view of the first main surface, the ground terminal is disposed between the first terminal and the second input/output terminal.

6. The acoustic wave filter according to claim **4**,

wherein the ground terminal is one of a plurality of ground terminals disposed on the first main surface,

wherein in plan view of the first main surface, a first virtual line connecting the first terminal and the second input/output terminal crosses a second virtual line connecting two ground terminals of the plurality of ground terminals, and

wherein the second virtual line is shorter than the first virtual line.

7. The acoustic wave filter according to claim **4**,

wherein the first substrate has piezoelectricity, and

wherein the acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices comprises an interdigital transducer electrode.

8. The acoustic wave filter according to claim **4**,

wherein the first substrate comprises silicon, and

wherein the acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices comprises a multilayer body having a first planar electrode, a piezoelectric thin film, and a second planar electrode in this order from the first main surface or the second main surface.

9. A radio frequency module comprising:

the acoustic wave filter according to claim **1**; and

a low-noise amplifier having an input terminal connected to the first input/output terminal.

10. The radio frequency module according to claim **9**,

wherein a capacitor is not disposed in series on a path connecting the first input/output terminal and an input port of an amplifying transistor included in the low-noise amplifier.

11. The radio frequency module according to claim **9**, further comprising:
a first filter connected to the input terminal; and
a mounting substrate on which the acoustic wave filter, the first filter, and the low-noise amplifier are disposed,
wherein a distance between the low-noise amplifier and the acoustic wave filter is longer than a distance between the low-noise amplifier and the first filter.

12. The radio frequency module according to claim **11**, further comprising:
a switch that has a common terminal, a first selection terminal, and a second selection terminal and that performs switching between a connection between the common terminal and the first selection terminal and a connection between the common terminal and the second selection terminal,
wherein the common terminal is connected to the input terminal,
wherein the first selection terminal is connected to the first input/output terminal, and
wherein the second selection terminal is connected to the first filter.

13. The acoustic wave filter according to claim **2**,
wherein a resonant frequency of the first serial arm resonant device is located at a frequency higher than or equal to the highest frequency end of a pass band of the acoustic wave filter.

14. The acoustic wave filter according to claim **2**, further comprising:
a first substrate having a first main surface and a second main surface facing each other,
wherein an acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices is formed on the first substrate, and
wherein the first input/output terminal, the second input/output terminal, a first terminal, and a ground terminal are disposed on the first main surface, the first terminal being connected to the first inductor and being disposed on the first path.

15. The acoustic wave filter according to claim **14**,
wherein in plan view of the first main surface, the ground terminal is disposed between the first terminal and the second input/output terminal.

16. The acoustic wave filter according to claim **14**,
wherein a plurality of ground terminals is disposed on the first main surface, the plurality of ground terminals including said ground terminal,
wherein in plan view of the first main surface, a first virtual line connecting the first terminal and the second input/output terminal crosses a second virtual line connecting two ground terminals of the plurality of ground terminals, and
wherein the second virtual line is shorter than the first virtual line.

17. The acoustic wave filter according to claim **14**,
wherein the first substrate has piezoelectricity, and
wherein the acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices comprises an interdigital transducer electrode.

18. The acoustic wave filter according to claim **14**,
wherein the first substrate comprises silicon, and
wherein the acoustic wave resonator included in each of the plurality of serial arm resonant devices and each of the one or more parallel arm resonant devices comprises a multilayer body having a first planar electrode, a piezoelectric thin film, and a second planar electrode in this order from the first main surface or the second main surface.

19. A radio frequency module comprising:
the acoustic wave filter according to claim **2**; and
a low-noise amplifier having an input terminal connected to the first input/output terminal.

20. The radio frequency module according to claim **19**,
wherein a capacitor is not disposed in series on a path connecting the first input/output terminal and an input port of an amplifying transistor included in the low-noise amplifier.

* * * * *